The pxCORE package

Core aggregation and key management for P3CTeX

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Abstract

pxCORE is the central aggregation package in the P3CTeX stack. It owns the pxCORE key family, which controls activation of standalone libraries (pxPRP, pxUML, pxSRC, ...) and offers a minimal LaTeX2e configuration API.

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1 Loading the package

In standalone documents, load pxCORE as a normal package:

```
\usepackage[<options>]{pxCORE}
```

In P3CTeX-based workflows, pxCORE is normally loaded by the document class (P3CTeX.cls) and configured via class options.

2 Readable tool selection

```
\usepackage[code,images,tables,diagrams,calculations,project-management]{pxCORE}
```

The keywords `code`, `images`, `tables` and `diagrams` select the existing LST, SRC, TAB and CMP tools. `diagrams-config` forwards diagram options, just like `diagrams-config`. `calculations` (or `CALC`) loads `pxCALC`; `project-management` (or `PM`) loads `pxPM`. The diagram entry point `pxDIAG` delegates to `pxDIAG`. All selection keys accept `=false`; a bare keyword means true. See the student-tools guide for examples and customisation.

3 Key family pxCORE

The `pxCORE` key family controls which libraries are activated and stores a few global flags:

`debug` (bool) enables internal debugging. When set, `\pxCOREIfDebugTF` will take its "true" branch.

`PRP` (bool) toggles loading of the `pxPRP` property-map library.

`UML` (bool) toggles loading of the `pxUML` UML diagram library.

`UMLconfig` (token list) raw option string passed to `pxUML`. When non-empty, `pxCORE` executes

```
\RequirePackage[<UMLconfig>]{pxUML}
```

instead of a bare `\RequirePackage{pxUML}`.

`SRC` (bool) toggles loading of the `pxSRC` included-sources library.

Keys are typically set via package options:

```
\usepackage[PRP,UML,SRC]{pxCORE}
```

Runtime debug settings use `\pxCOREsetup` (see below).

4 User command reference

4.1 \pxCOREsetup

Synopsis

```
\pxCOREsetup{<key>=<value>,...}
```

Updates runtime debug settings after package loading. This is useful to toggle flags such as `debug` programmatically or to adjust options in local scopes.

Example:

```
\pxCOREsetup{debug=true}
...
\pxCOREsetup{debug=false}
```

4.2 \pxCOREIfDebugTF

Synopsis

```
\pxCOREIfDebugTF{<true-code>}{<false-code>}
```

Branch on the `debug` flag in the `pxCORE` key family. Both branches are arbitrary LaTeX code.

Example:

```
\pxCOREIfDebugTF
  {\typeout{[pxCORE] Debug mode enabled}}
  {}
```

5 Minimal example

```
\usepackage[PRP,UML,SRC]{pxCORE}
\begin{document}
\pxCOREsetup{debug=true}
\pxCOREIfDebugTF{[DEBUG-ON]}{[DEBUG-OFF]}
\section{Configuration diagnostics}
Unknown tool keys produce an error instead of being ignored. Runtime
\verb|\pxCOREsetup| accepts \texttt{debug};
it rejects module selection and load-time configuration. Select tools in the
class/package options and customise loaded tools through their own setup commands.
\end{document}
```

6 Configuration diagnostics

Unknown tool keys produce an error instead of being ignored. Runtime `\pxCOREsetup` accepts `debug`; it rejects module selection and load-time configuration. Select tools in the class/package options and customise loaded tools through their own setup commands.